



INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO



Introduction to Cryptography

Lab 09: Mode of operation CTR

April 30, 2026

Please do the following exercises and write the pseudocode by hand in your notebook or tablet. Upload a photo or image to Microsoft Teams.

1. Pseudocode

1. Write an algorithm to encipher a plaintext $M \in \{0,1\}^*$ using the mode of operation CTR and the toy block cipher that you have been developed before.
2. Write an algorithm to decipher a ciphertext C using the mode of operation CTR within the toy block cipher to recover the plaintext M .

2. Programming exercises

Do the following exercises by your own, using C/C++ programming language and bitwise operations.

1. Design and implement a function for each of the two algorithms that you developed in Section 1. Your implementation must satisfy the following:
 - To store the counter you will use two unsigned char variables C_0, C_1 . In C_0 you will store a randomly generated integer. Initialize $C_1 = 0$, then encipher C_0, C_1 with your TBC and use the output to encipher the first block of the plaintext M . To encipher the next block of M just increment C_1 and repeat the process.
 - At the beginning of the ciphertext C store the values of C_0 .
 - Print the ciphertext in hexadecimal.